Question ID: 14360f84

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Hard

Question

$$4x - 9y = 9y + 5$$

$$hy = 2 + 4x$$

In the given system of equations, h is a constant. If the system has no solution, what is the value of h?

Answer

A. -9

B. 0

C. 9

D. 18

Correct Answer: D

Rationale

Choice D is correct. A system of two linear equations in two variables, x and y, has no solution if the lines represented by the equations in the *xy*-plane are distinct and parallel. The graphs of two lines in the *xy*-plane represented by equations in the form $Ax + By = C$, where A, B, and C are constants, are parallel if the coefficients for x and y in one equation are proportional to the corresponding coefficients in the other equation. The first equation in the given system can be written in the form $Ax + By = C$ by subtracting $9y$ from both sides of the equation to yield $4x - 18y = 5$. The second equation in the given system can be written in the form $Ax + By = C$ by subtracting $4x$ from both sides of the equation to yield $-4x + hy = 2$. The coefficient of x in this second equation, -4, is -1 times the coefficient of x in the first equation, 4. For the lines to be parallel, the coefficient of y in the second equation, h, must also be -1 times the coefficient of y in the first equation, -18. Thus, $h = -1(-18)$, or $h = 18$. Therefore, if the given system has no solution, the value of h is 18.

Choice A is incorrect. If the value of h is -9, then the given system would have one solution, rather than no solution.

Choice B is incorrect. If the value of h is 0, then the given system would have one solution, rather than no solution.

Choice C is incorrect. If the value of h is 9, then the given system would have one solution, rather than no solution.